

MySQL Assignment 2 – Querying Data

Objective

Populate the Employee Database built in Assignment 1 with real records, then write SQL queries to filter, sort, aggregate, group, and join that data to answer practical business questions.

Database Populated

13 departments • 4 locations (Chennai, Bangalore, Hyderabad, Pune) • 30 employees, spanning hire dates from 2015 to 2019 and salaries from ₹45,000 to ₹100,000.

Tasks Completed

- Distinct Values – retrieved unique salary figures.
- Alias (AS) – relabelled age and salary columns for readability.
- Where Clause & Operators – filtered by salary and hire date; located and filled a missing designation.
- Order By – sorted employees by department then salary.
- Limit – pulled the first 5 employees hired in 2018.
- Aggregate Functions – summed Finance salaries; found the minimum employee age.
- Group By – max salary per location; average salary per Analyst-type role.
- Having – departments understaffed (<3 employees); locations with average female age under 30.
- Inner Join – employee, designation, and department in one result.
- Left Join – employee counts per department, including departments with none.
- Right Join – employees per location, including locations with none assigned.

Tools Used

MySQL 8.0, MySQL Workbench

Outcome

All 11 querying tasks from the assignment brief were completed and verified against the populated dataset, covering the full range of SQL retrieval concepts: filtering, sorting, aliasing, aggregation, conditional grouping, and all three major join types.